

```

1  clc;
2  clear;
3  close all;
4
5  % Input Data
6  P = [0 0 1 1;
7       0 1 0 1];
8
9  % Target Data
10 T = [0 1 1 1];
11
12 % Create Network
13 net = feedforwardnet(5);
14
15 % Use sigmoid in output layer
16 net.layers{2}.transferFcn = 'logsig';
17
18 % Train Network
19 [net,tr] = train(net,P,T);
20
21 % Test Network
22 Y = net(P);
23
24 disp('Input Data');
25 disp(P);
26
27 disp('Target Output');
28 disp(T);
29
30 disp('Network Output');
31 disp(Y);
32
33 disp('Binary Output');
34 disp(round(Y));
35
36 % Performance Plot
37 figure;
38 plotperform(tr);

```

```

Input Data
    0    0    1    1
    0    1    0    1

Target Output
    0    1    1    1

Network Output
    0.6974    0.5109    0.9075    0.9310

Binary Output
    1    1    1    1

>>

```

Network Diagram

Training Results

Training finished: Met validation criterion ✔

Training Progress

Unit	Initial Value	Stopped Value	Target Value	
Epoch	0	6	1000	▲
Elapsed Time	-	00:00:00	-	
Performance	0.124	4.02e-08	0	
Gradient	0.0182	1.87e-07	1e-07	
Mu	0.001	1e-09	1e+10	
Validation Checks	0	6	6	▼

Training Algorithms

Data Division: Random dividerand
Training: Levenberg-Marquardt trainlm
Performance: Mean Squared Error mse
Calculations: MEX

Training Plots

Performance

Training State

Error Histogram

Regression





